CS343 Project 2 – Database Report

1. Database Design Overview

The application uses SQLite through Prisma ORM. The database stores users and sessions, hackathon groups, teams and invitations, programming problems and test cases, team files, and graded submissions. The design uses foreign keys and junction tables to represent one-to-many and many-to-many relationships while avoiding duplicated data.

Area	Relations	Purpose
Accounts	User, Session	Accounts and sign-in sessions (there is a 30-min inactivity rule)
Groups	Group, GroupMember, GroupInvite	A hackathon event, its single admin, its settings, members and pending invitations
Teams	Team, TeamMembers, TeamInvite	Teams within ta group, their leader, members and pending invitations.
Problems	Problem, TestCase	The ordered problem sequence of a group and the test cases of each problem.
Shared editor	Codebase, odebaseFile	One codebase per team per problem: its folder tree and the contents of each file.
Grading	Submission, SubmissionResult	Each submission of a team for a problem, its score (P, T) and the outcome of each test case

Design descisions:

- Many to many relationships use junction tables. A user can belong to many groups and many teams, so membership is stored in GroupMember and TeamMember, each with a composite unique key (groupId, userId) / (teamId, userId) so a user can't be added twice.
- Values that can be computed are not stored. The end time of an event is startedAt + durationMin and is computed when needed. Leaderboard positions and final scores are computed from Submission rows on request, so they can never be outdated.
- Test case data is not copied into results. SubmissionResult points at its TestCase; in input and expected output are read from there. Only the program's actual output is stored and only for failed variable tests, so hidden outputs are never kept.
- Referential integrity is enforced by foreign keys with ON DELETE CASCADE where the child cannot exist without its parent and SET NULL for Team.activeProblemId.

2. Entity-Relationship Diagram

The ER diagram below reflects the current Prisma schema used by the project.

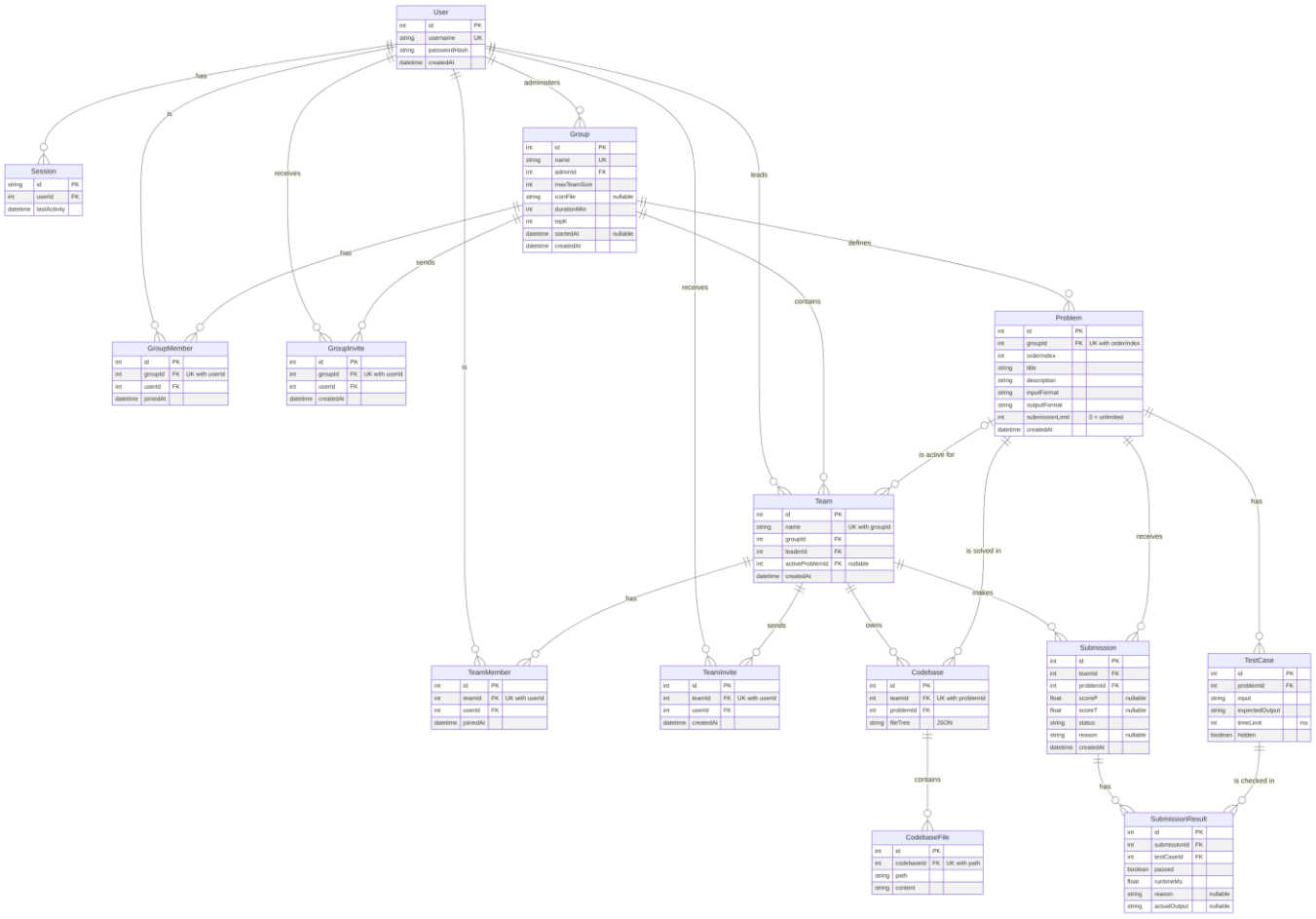


Figure 1: Entity-Relationship Diagram

3. Main Relationships

- A User can have many Sessions, administer Groups, lead Teams, and belong to Groups and Teams through GroupMember and TeamMember.
- A Group contains many Teams, Problems, GroupMembers and GroupInvites.
- A Team belongs to one Group, has one leader, and can have many TeamMembers, TeamInvites, TeamFiles and Submissions.
- A Problem belongs to one Group and has many TestCases and Submissions.
- A Submission belongs to exactly one Team and one Problem and stores the calculated leaderboard values scoreP and scoreT.

4. Normalization

All relations are designed to satisfy Third Normal Form (3NF). Each table has a primary key, attributes contain atomic values, non-key attributes depend on the whole key, and there are no unnecessary transitive dependencies between non-key attributes. The table below summarises this.

Relation	Normal Form	Reason
User	3NF	id is the primary key; username is unique; all other attributes describe only that user.
Session	3NF	Each session is identified by id and stores only its user reference and last activity time.
Group	3NF	All settings depend directly on the group id; adminId is a foreign key to User.
GroupMember	3NF	Junction relation between User and Group; the unique (groupId, userId) pair prevents duplicate membership.

Team	3NF	Team attributes depend on the team id; groupId and leaderId reference Group and User.
TeamMember	3NF	Junction relation between User and Team; no non-key attribute depends on another non-key attribute.
TeamInvite	3NF	Stores one pending User–Team invitation; (teamId, userId) is unique.
GroupInvite	3NF	Stores one pending User–Group invitation; (groupId, userId) is unique.
TeamFile	3NF	Each file belongs to one team; path and content depend on the file record, with (teamId, path) unique.
Problem	3NF	Problem details depend on id; groupId links the problem to its group and orderIndex defines its order within that group.
TestCase	3NF	Each test case belongs to one problem and its input, expected output, time limit and hidden flag depend on that test case.
Submission	3NF	Each submission links one team to one problem; scoreP, scoreT, status and createdAt describe that submission only.

5. Conclusion

The database design separates users, memberships, invitations, teams, problems, test cases, files and submissions into focused relations. Foreign keys preserve referential integrity, while unique constraints prevent duplicate memberships, invitations, team names within a group, problem order positions, and team file paths. This keeps the schema simple, consistent and suitable for the features required by the hackathon platform.